

AIAC ASSIGNMENT 7.5

NAME: ISHA MADEEHA

HALLTICKET NO: 2303A51017

BATCH-01

Task 1 (Mutable Default Argument – Function Bug)

Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

Bug: Mutable default argument

```
def add_item(item, items=[]):
```

```
    items.append(item)
```

```
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.

#before fixing

```
def add_item(item, items=[]):
```

```
    items.append(item)
```

```
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```

#after fixing

```
def add_item(item, items=None):
```

```
    if items is None:
```

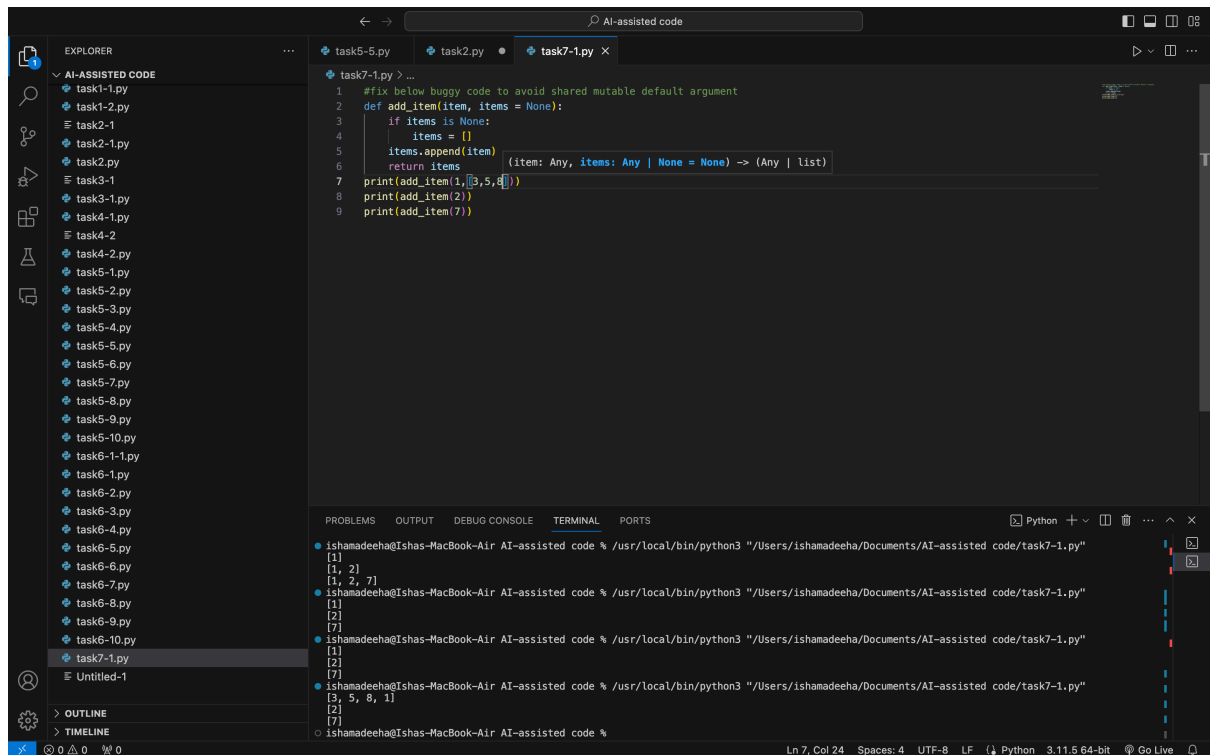
```
        items = []
```

```
    items.append(item)
```

```
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```



Task 2 (Floating-Point Precision Error)

Task: Analyze given code where floating-point comparison fails.

Use AI to correct with tolerance.

Bug: Floating point precision issue

```
def check_sum():
```

```
    return (0.1 + 0.2) == 0.3
```

```
    print(check_sum())
```

Expected Output: Corrected function

#befor fixing

```
def check_sum():
```

```
    return (0.1 + 0.2) == 0.3
```

```
    print(check_sum())
```

#after fixing

```
import math
```

```
def check_sum():
```

```
    return math.isclose(0.1 + 0.2, 0.3)
```

The screenshot shows a VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files task1-1.py through task7-2.py. The main editor window displays the code for task7-2.py:

```
def check_sum():
    return abs((0.1+0.2) - 0.3) < 1e-10
print(check_sum())
```

The terminal window at the bottom shows the command to run the script and the output:

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py"
False
ishamadeeha@Ishas-MacBook-Air AI-assisted code % 0.1+0.4
zsh: command not found: 0.1+0.4
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py"
Traceback (most recent call last):
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py", line 3, in <module>
    print(check_sum())
          ~~~~~^~~~~
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py", line 2, in check_sum
    return abs(0.1 + 0.2, 1) == 0.3
            ~~~~~^~~~~~
TypeError: abs() takes exactly one argument (2 given)
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py"
True
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py"
Traceback (most recent call last):
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py", line 3, in <module>
    PF1
    ^
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py", line 2, in check_sum
    return abs(0.1+0.2, 1) == 0.3
            ~~~~~^~~~~~
TypeError: abs() takes exactly one argument (2 given)
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-2.py"
True
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

Task: Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix.

```
def countdown(n):
```

```
return countdown(n-1)
```

Expected Output : Correct recursion with stopping condition.

```
def countdown(n):
```

```
return countdown(n-1)
```

```
countdown(5)
```

#after fixing

```
def countdown(n):
```

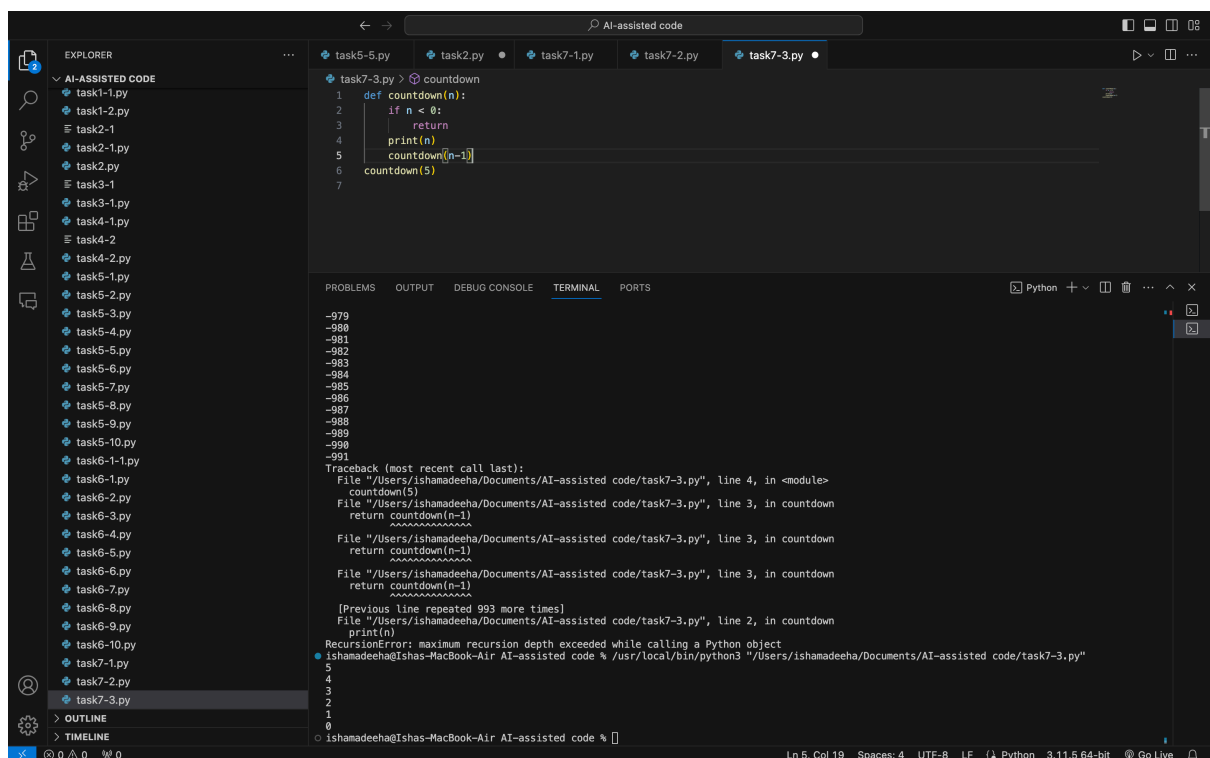
```
    if n < 0:
```

```
        return
```

```
    print(n)
```

```
    return countdown(n-1)
```

```
countdown(5)
```



Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error. Use AI to fix it.

Bug: Accessing non-existing key

```
def get_value():
```

```
    data = {"a": 1, "b": 2}
```

```
    return data["c"]
```

```
print(get_value())
```

Expected Output: Corrected with .get() or error handling.

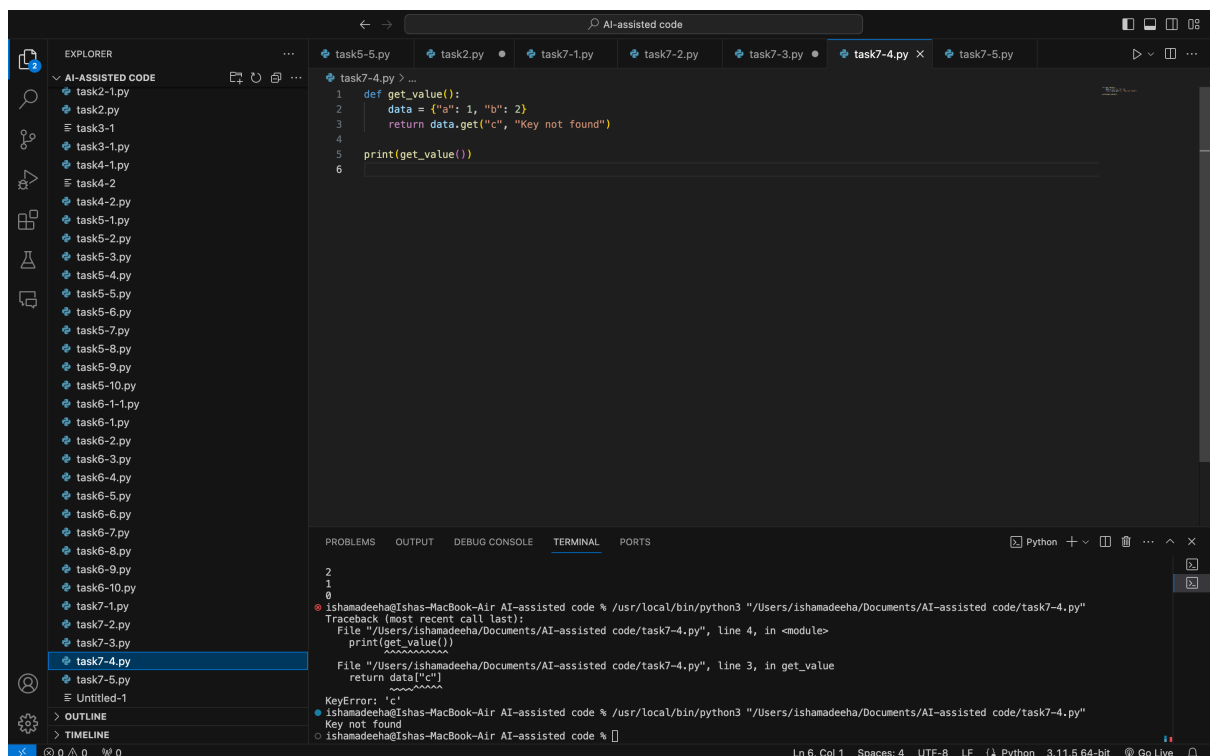
#before fixing

```
def get_value():  
    data = {"a": 1, "b": 2}  
    return data["c"]  
    print(get_value())
```

#after fixing

missing dictionary to accession non existing key

```
def get_value():  
    data = {"a": 1, "b": 2, "c": 3}  
    return data.get("c", None)  
    print(get_value())
```



Task 5 (Infinite Loop – Wrong Condition)

Task: Analyze given code where loop never ends. Use AI to detect and fix it.

Bug: Infinite loop

```
def loop_example():
```

```
i = 0
```

```
while i < 5:
```

```
print(i)
```

Expected Output: Corrected loop increments i

#before fixing

```
def loop_example():
```

```
i = 0
```

```
while i < 5:
```

```
print(i)
```

#after fixing

```
def loop_example():
```

```
    i = 0
```

```
    while i < 5:
```

```
        print(i)
```

```
        i += 1 # Increment i to avoid infinite loop
```

```
loop_example() # Call the function to execute the loop
```

The screenshot shows a VS Code editor with a file explorer on the left containing a directory of files named 'task2-1.py' through 'task7-5.py'. The main editor window displays a Python script named 'task7-5.py' with the following code:

```
1 def loop_example():
2     i = 0
3     while i < 5:
4         print(i)
5         i += 1 # Increment i
6
7 loop_example()
8
```

Below the code editor, the 'TERMINAL' panel shows the output of running the script. It displays a 'KeyError: 'c'' message, followed by a list of commands executed in the terminal, all of which are identical: `usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-5.py"`. The output shows the numbers 0, 1, 2, 3, and 4, which are the values of `i` during the loop iterations.

Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to fix it.

Bug: Wrong unpacking

```
a, b = (1, 2, 3)
```

Expected Output: Correct unpacking or using `_` for extra values.

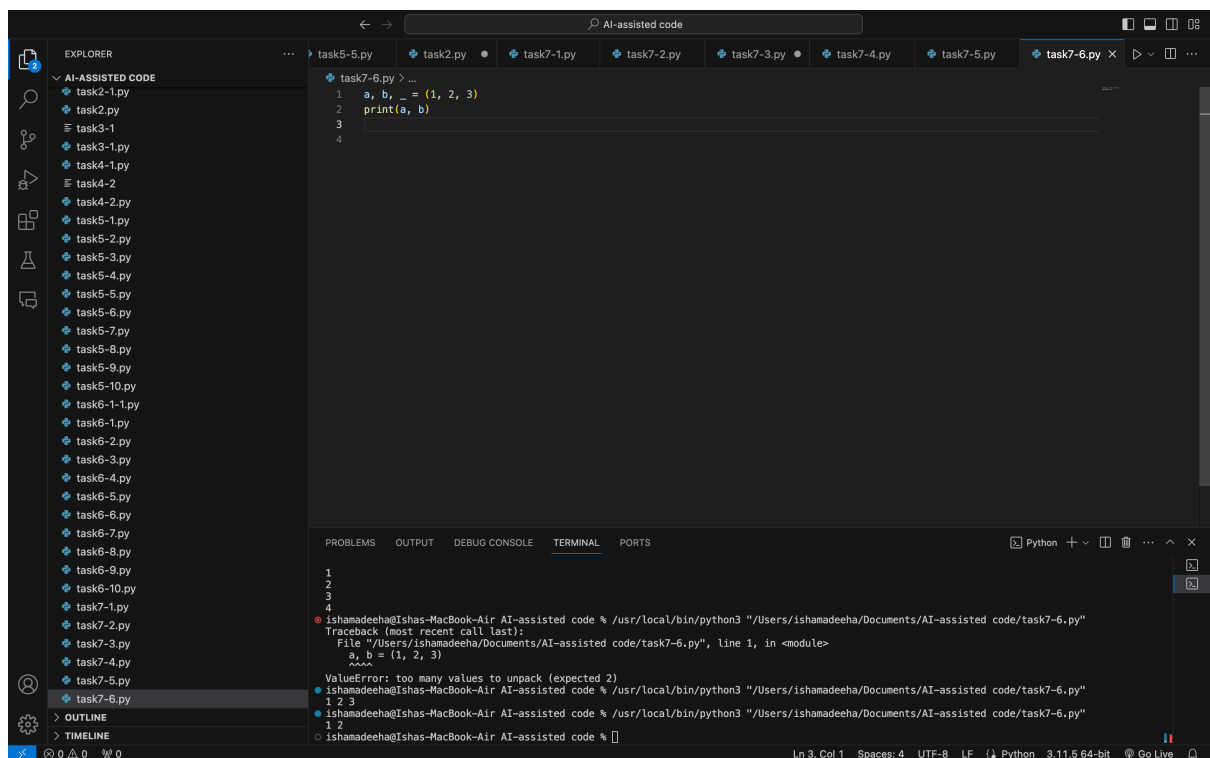
#after fixing

Task: Analyze given code where tuple unpacking fails. Use AI to fix it.

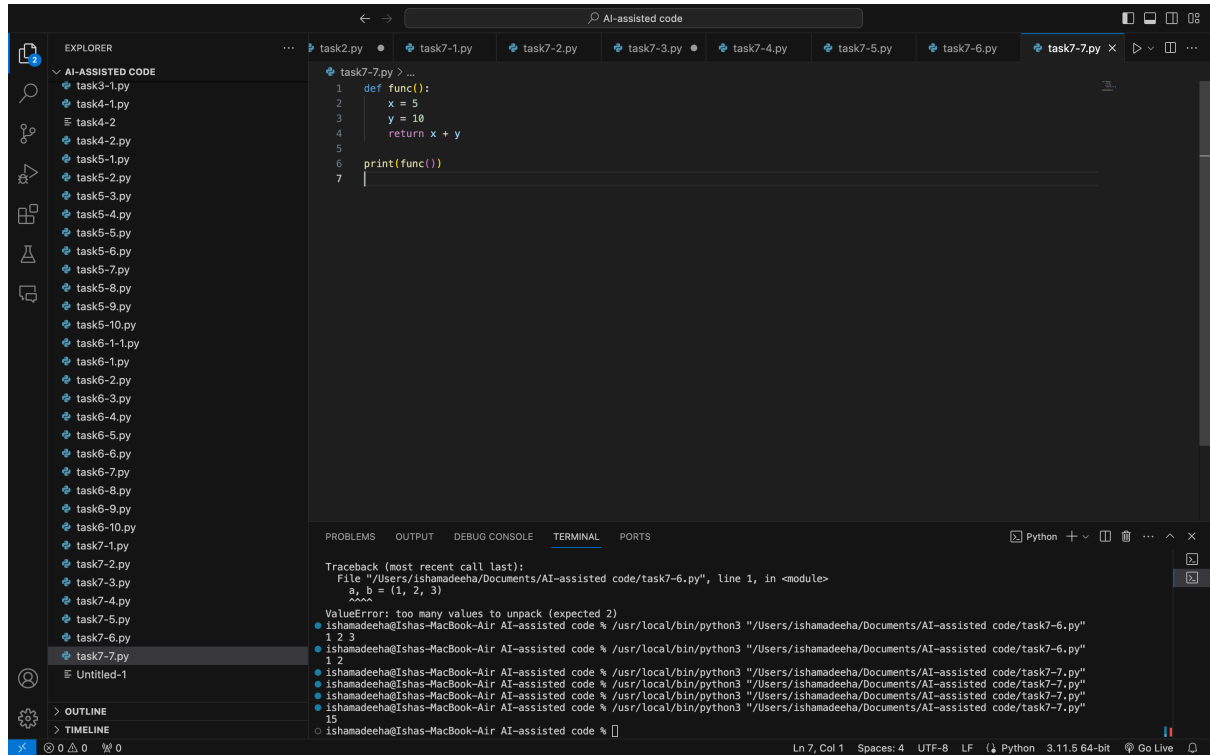
Bug: Wrong unpacking - fixed by using underscore for extra values

```
a, b, _ = (1, 2, 3)
```

```
print(f"a = {a}, b = {b}")
```



Task-7:



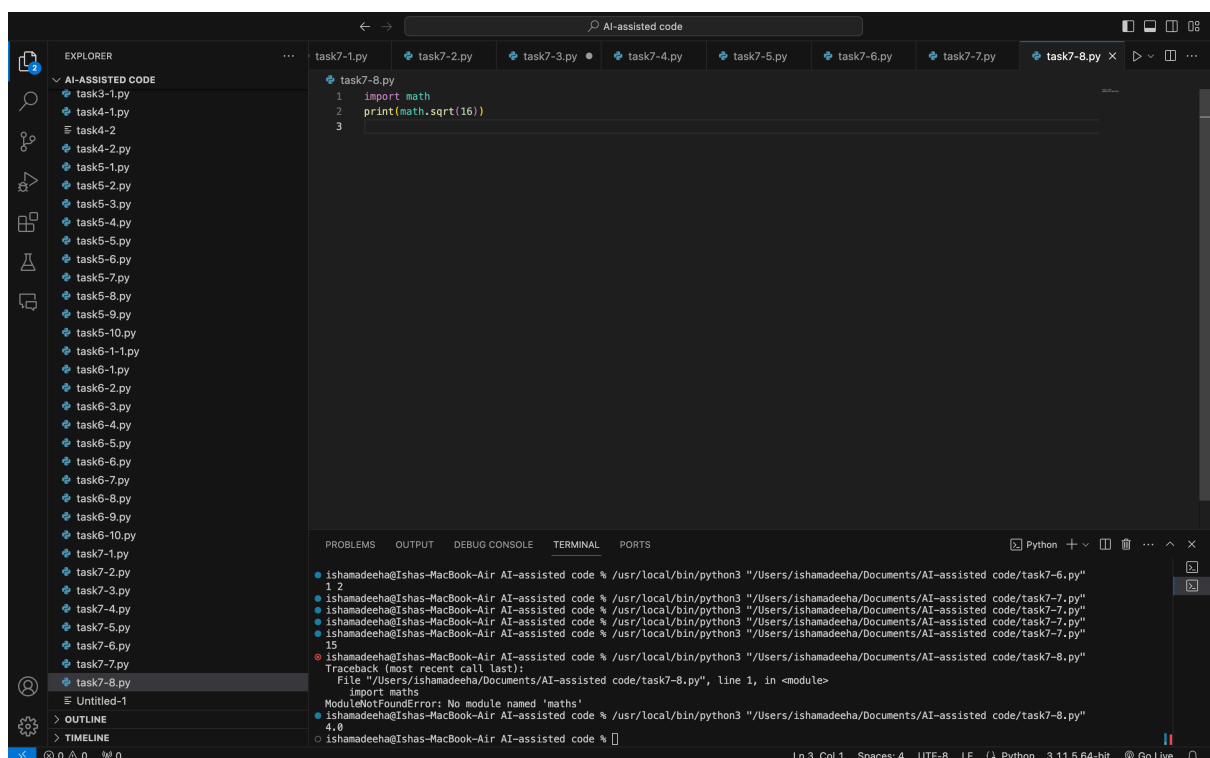
The screenshot shows the VS Code editor interface. The Explorer panel on the left lists files from task3-1.py to task7-10.py. The main editor window displays task7-7.py with the following code:

```
1 def func():  
2     x = 5  
3     y = 10  
4     return x + y  
5  
6 print(func())  
7
```

The Terminal panel at the bottom shows the following output:

```
Traceback (most recent call last):  
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-6.py", line 1, in <module>  
    a, b = (1, 2, 3)  
ValueError: too many values to unpack (expected 2)  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-6.py"  
1 2 3  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
1 2  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
15  
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

Task - 8:



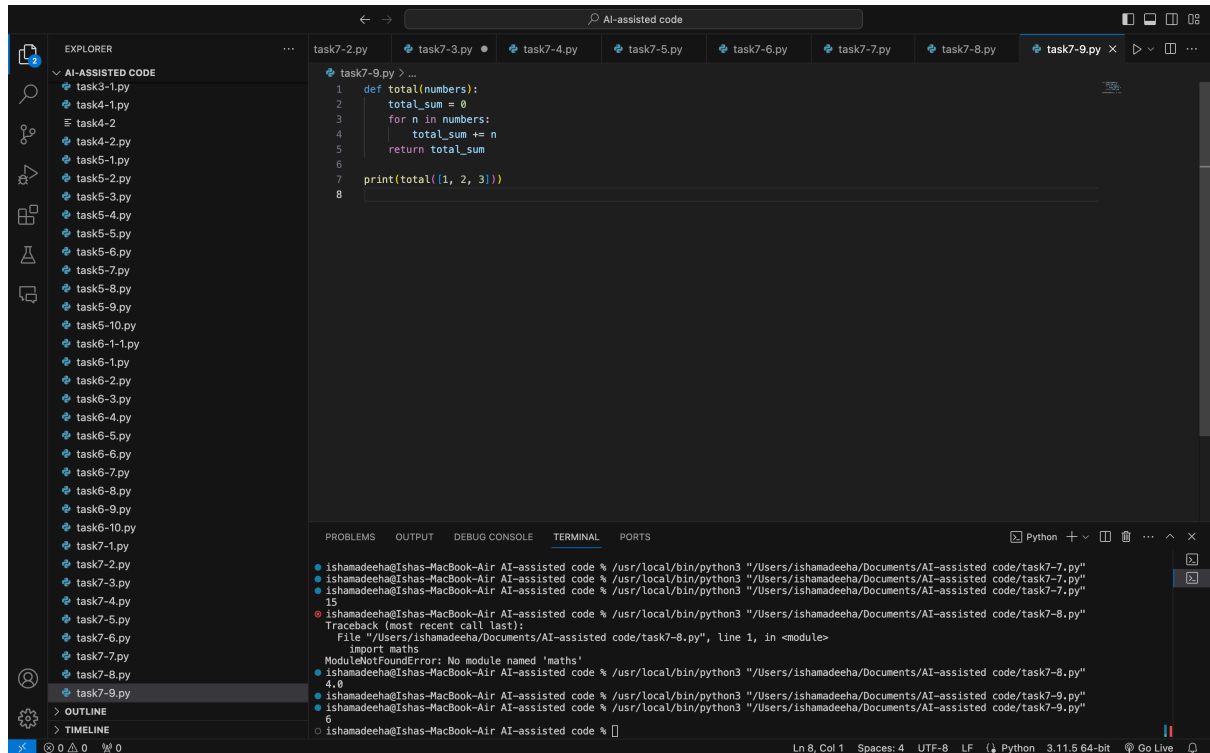
The screenshot shows the VS Code editor interface. The Explorer panel on the left lists files from task3-1.py to task7-10.py. The main editor window displays task7-8.py with the following code:

```
1 import math  
2 print(math.sqrt(16))  
3
```

The Terminal panel at the bottom shows the following output:

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-6.py"  
1 2  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-7.py"  
15  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-8.py"  
Traceback (most recent call last):  
  File "/Users/ishamadeeha/Documents/AI-assisted code/task7-8.py", line 1, in <module>  
    import math  
ModuleNotFoundError: No module named 'maths'  
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-8.py"  
4.0  
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```


Task - 9:

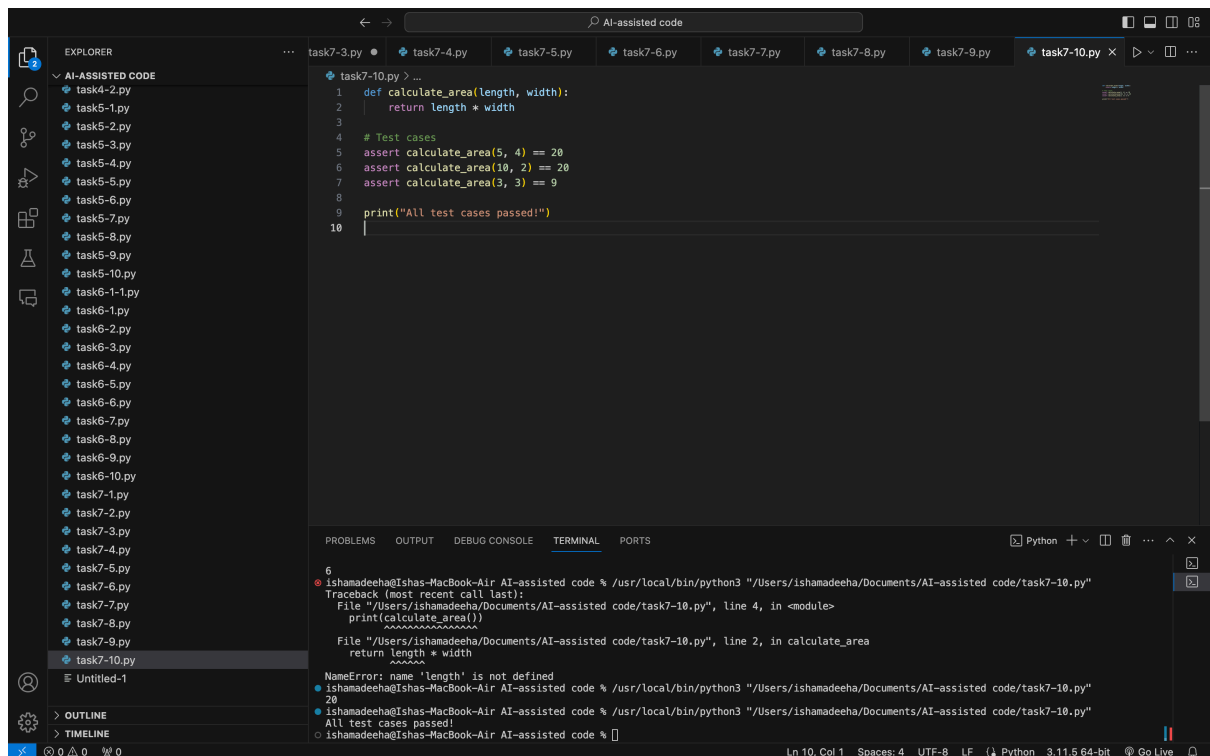


The screenshot shows the VS Code editor interface. The Explorer panel on the left displays a list of files under 'AI-ASSISTED CODE', including task3-1.py through task7-9.py. The main editor window shows the code for task7-9.py, which defines a function `total(numbers)` that calculates the sum of elements in a list. The code is as follows:

```
1 def total(numbers):
2     total_sum = 0
3     for n in numbers:
4         total_sum += n
5     return total_sum
6
7 print(total([1, 2, 3]))
8
```

The terminal panel at the bottom shows the output of running the code. It displays a series of error messages from the AI-assisted code, including a `ModuleNotFoundError: No module named 'maths'` and a `Traceback (most recent call last):` error. The status bar at the bottom indicates the file is at line 8, column 1, with 4 spaces, in UTF-8 encoding, using Python 3.11.5 64-bit.

Task - 10:



The screenshot shows the VS Code editor interface. The Explorer panel on the left displays a list of files under 'AI-ASSISTED CODE', including task4-2.py through task7-10.py. The main editor window shows the code for task7-10.py, which defines a function `calculate_area(length, width)` that calculates the area of a rectangle. The code is as follows:

```
1 def calculate_area(length, width):
2     return length * width
3
4 # Test cases
5 assert calculate_area(5, 4) == 20
6 assert calculate_area(10, 2) == 20
7 assert calculate_area(3, 3) == 9
8
9 print("All test cases passed!")
10
```

The terminal panel at the bottom shows the output of running the code. It displays a series of error messages from the AI-assisted code, including a `NameError: name 'length' is not defined` and a `Traceback (most recent call last):` error. The status bar at the bottom indicates the file is at line 10, column 1, with 4 spaces, in UTF-8 encoding, using Python 3.11.5 64-bit.

Task - 11:

```
task7-11.py > ...
1 def add_values(value):
2     return 5 + int(value)
3
4 # Test cases
5 assert add_values("10") == 15
6 assert add_values("0") == 5
7 assert add_values("25") == 30
8
9 print("All test cases passed!")
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + - - - - -

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-11.py"

Traceback (most recent call last):

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-11.py", line 4, in <module>

print(add_values())

~~~~~

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-11.py", line 2, in add\_values

return 5 + "10"

~~~~~

TypeError: unsupported operand type(s) for +: 'int' and 'str'

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-11.py"

15

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-11.py"

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 10, Col 1 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Task - 12:

```
task7-12.py > ...
1 def combine(numbers):
2     return "Numbers: " + " ".join(map(str, numbers))
3
4 # Test cases
5 assert combine([1, 2, 3]) == "Numbers: 1 2 3"
6 assert combine([4, 5]) == "Numbers: 4 5"
7 assert combine([10]) == "Numbers: 10"
8
9 print("All test cases passed!")
10
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + - - - - -

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-12.py"

Traceback (most recent call last):

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-12.py", line 4, in <module>

print(combine())

~~~~~

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-12.py", line 2, in combine

return "Numbers: " + [1, 2, 3]

~~~~~

TypeError: can only concatenate str (not "list") to str

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-12.py"

Numbers: [1, 2, 3]

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-12.py"

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 10, Col 1 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Task - 13:

```
task7-13.py > ...
1 def repeat_text(times):
2     return "Hello" * int(times)
3
4 # Test cases
5 assert repeat_text(2.5) == "HelloHello"
6 assert repeat_text(3.0) == "HelloHelloHello"
7 assert repeat_text(1.9) == "Hello"
8
9 print("All test cases passed!")
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-13.py"

Traceback (most recent call last):

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-13.py", line 4, in <module>

print(repeat_text())

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-13.py", line 2, in repeat_text

return "Hello" * 2.5

TypeError: can't multiply sequence by non-int of type 'float'

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-13.py"

HelloHello

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-13.py"

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 10, Col 1 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Task - 14:

```
task7-14.py > ...
1 def compute(value=None):
2     if value is None:
3         value = 0
4     return value + 10
5
6 # Test cases
7 assert compute() == 10
8 assert compute(5) == 15
9 assert compute(20) == 30
10
11 print("All test cases passed!")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py"

Traceback (most recent call last):

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py", line 5, in <module>

print(compute())

File "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py", line 3, in compute

return value + 10

TypeError: unsupported operand type(s) for +=: 'NoneType' and 'int'

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py"

10

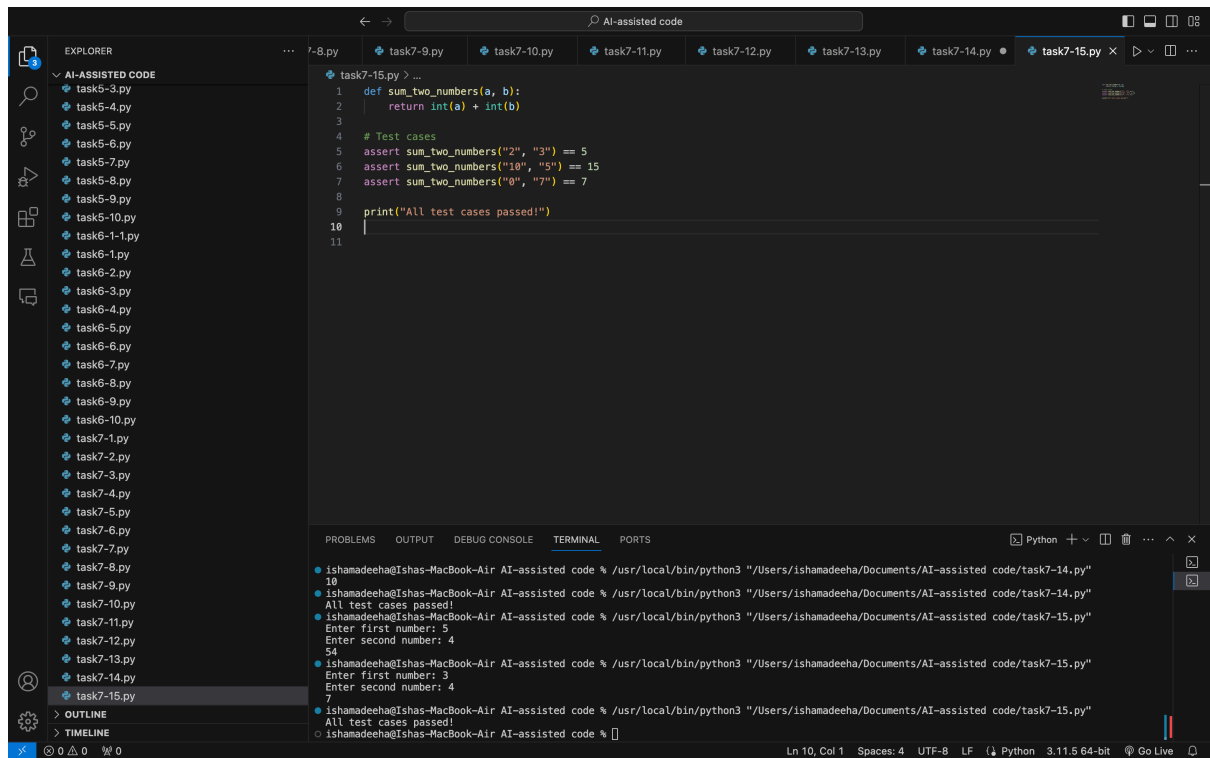
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py"

All test cases passed!

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 11, Col 32 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Task - 15:



The screenshot displays the Visual Studio Code interface. The Explorer sidebar on the left shows a project named 'AI-ASSISTED CODE' containing numerous files, including 'task7-15.py' which is currently selected. The main editor window shows the code for 'task7-15.py'.

```
1 def sum_two_numbers(a, b):
2     return int(a) + int(b)
3
4 # Test cases
5 assert sum_two_numbers("2", "3") == 5
6 assert sum_two_numbers("10", "5") == 15
7 assert sum_two_numbers("0", "7") == 7
8
9 print("All test cases passed!")
10
11
```

Below the editor, the 'TERMINAL' panel is active, showing the execution of the script. The output indicates that all test cases passed for three different inputs: (2, 3), (10, 5), and (0, 7).

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py"
10
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-14.py"
All test cases passed!
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-15.py"
Enter first number: 5
Enter second number: 4
54
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-15.py"
Enter first number: 3
Enter second number: 4
7
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task7-15.py"
All test cases passed!
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

The status bar at the bottom indicates the current cursor position is at line 10, column 1, with 4 spaces, in UTF-8 encoding, using the Python interpreter (3.11.5 64-bit).